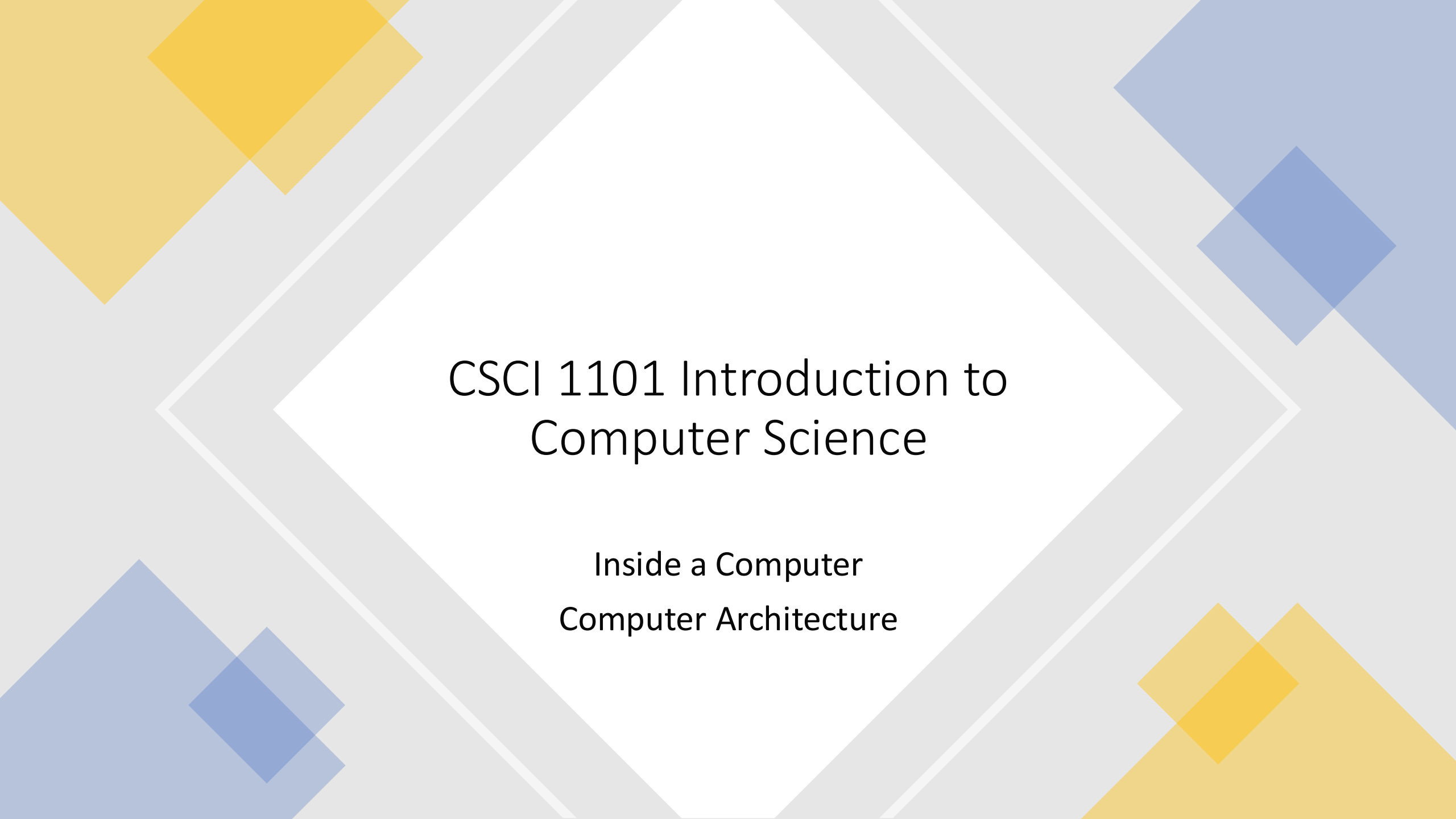


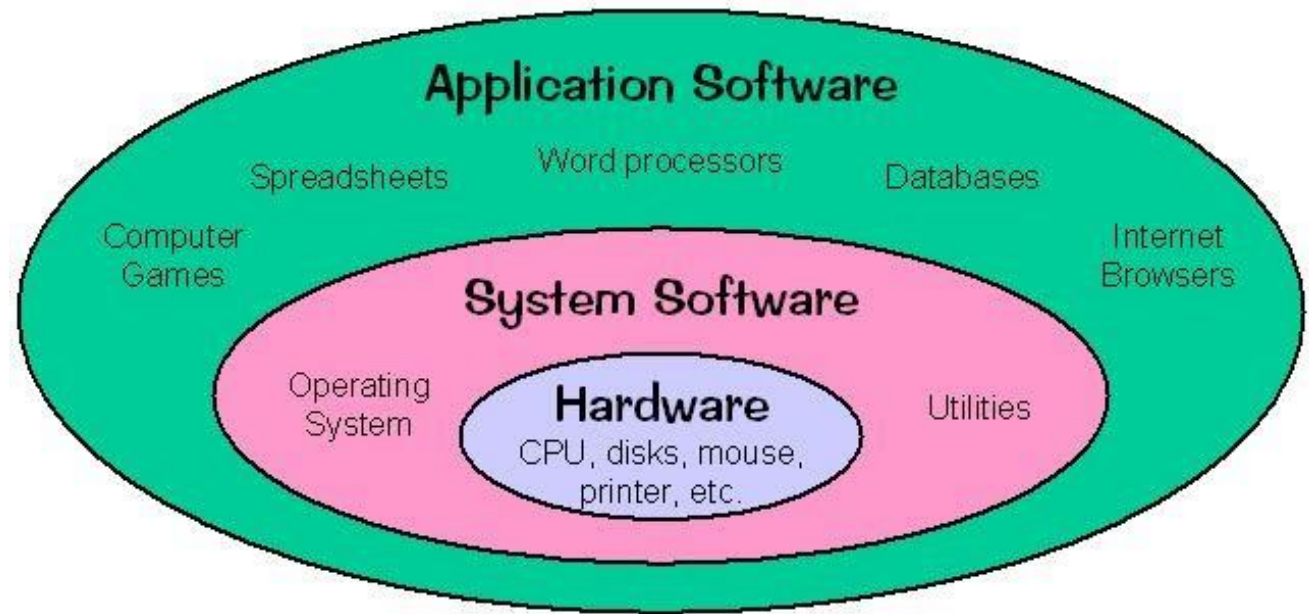


CSCI 1101 Introduction to Computer Science

Inside a Computer
Computer Architecture

Inside a Computer

- Software
 - OS (Operating System)
 - Applications (browser, Excel, etc.)
- Hardware
 - CPU
 - Mouse
 - Keyboard



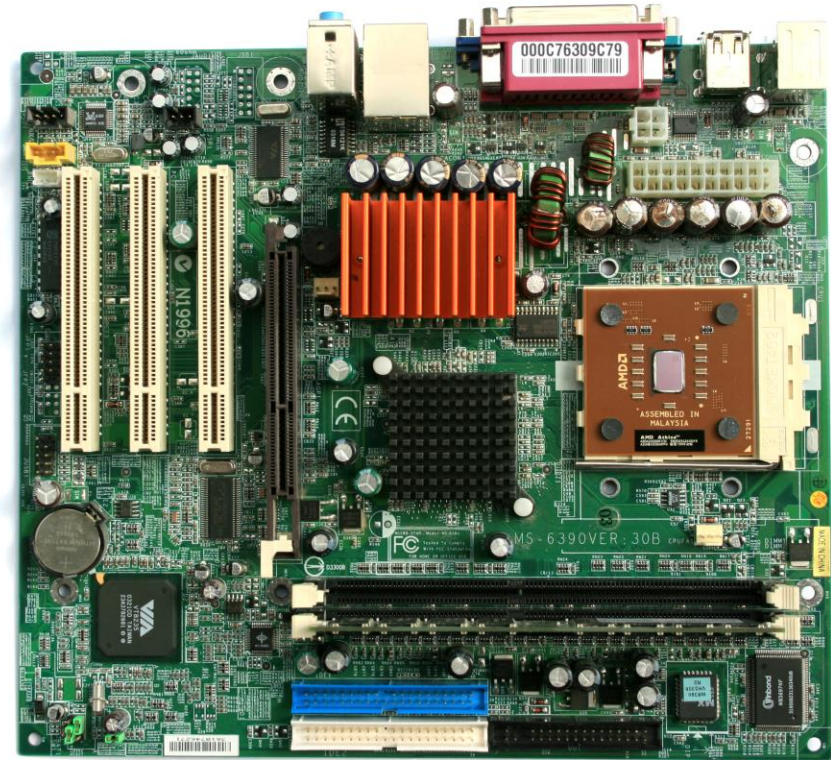
Hardware Components

- Motherboard
- Disk Storage (HDD, SSD)
- RAM (Main Memory)
- CPU
- Buses (USB, Keyboard, Screen, etc.)



Motherboard

- "Mother" of all components attached
- Responsible for the following
 - Allocating Power
 - Communication between hardware components
 - Synchronizing components
- Contains firmware or BIOS
- <https://sketchfab.com/3d-models/msi-b85-g41-motherboard-61b20aee38a44d00908e851e6f6ea240>



Disk Storage (Hard Drive)

- Can be Internal or External
- Types of storage
 - Hard Disk Drive (HDD) - uses magnetic discs and arm needle
 - Solid-State Drive (SSD) - uses non-volatile memory similar to USB
- Where all data is stored
 - Operating System
 - Applications
 - Files
- Data remains after power off



HDD – uses magnetic discs

RAM (Main Memory)

- Random Access Memory (RAM)
- Stores data and applications currently in use
- Main Memory must be
 - Easy to read
 - Easy to write
 - Able to access any memory address at any time
- OS controls when memory should be loaded or removed
- Data is lost after power off



CPU

- Central Processing Unit (CPU)
- Handles all instructions received from hardware and software
- Main components
 - ALU (Arithmetic Logic Unit) - performs mathematical, logical, decision operations
 - CU (Control Unit) - directs all processor operations
- CPU types
 - 32-bit architecture
 - 64-bit architecture



Buses

- Used to send data or power bidirectionally to various hardware components
- Types
 - Internal – Connect CPU, RAM, fan, etc.
 - External – Connect keyboard, mouse, monitor, etc.

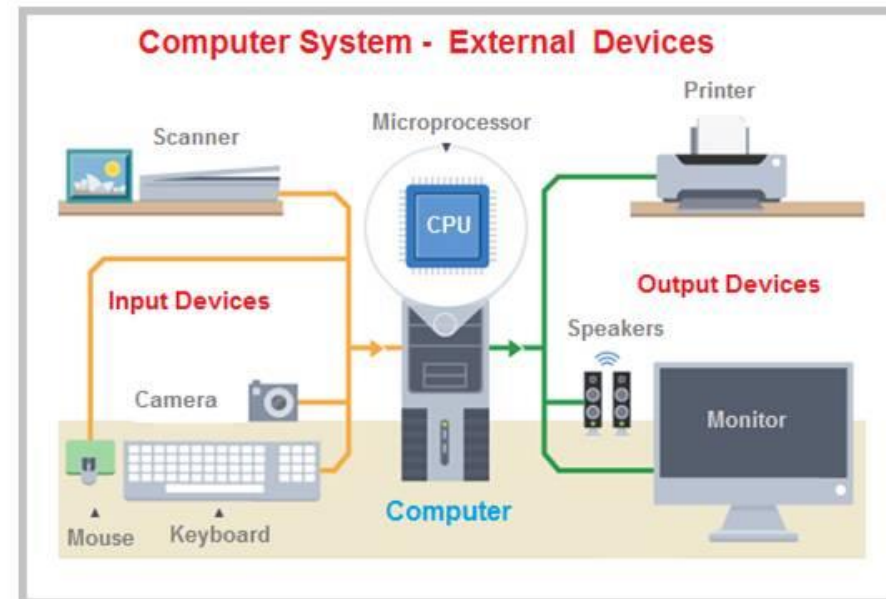
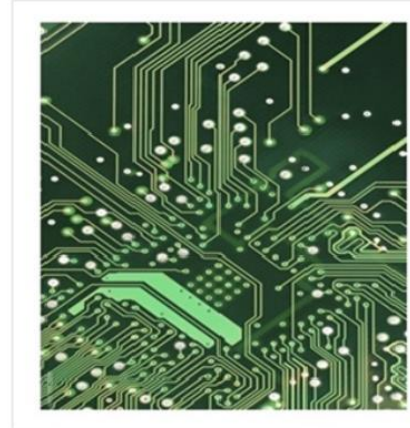
Cable Buses

Buses - Connection Cables

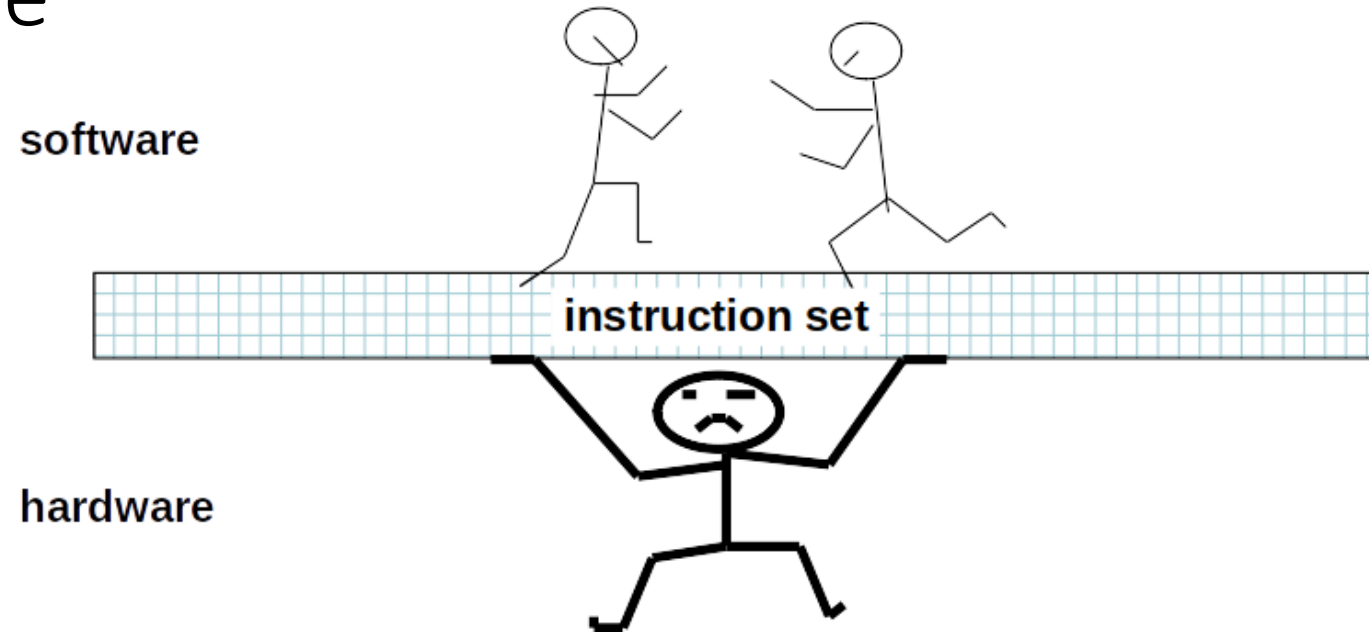


Motherboard Buses

Buses - Motherboard Buses



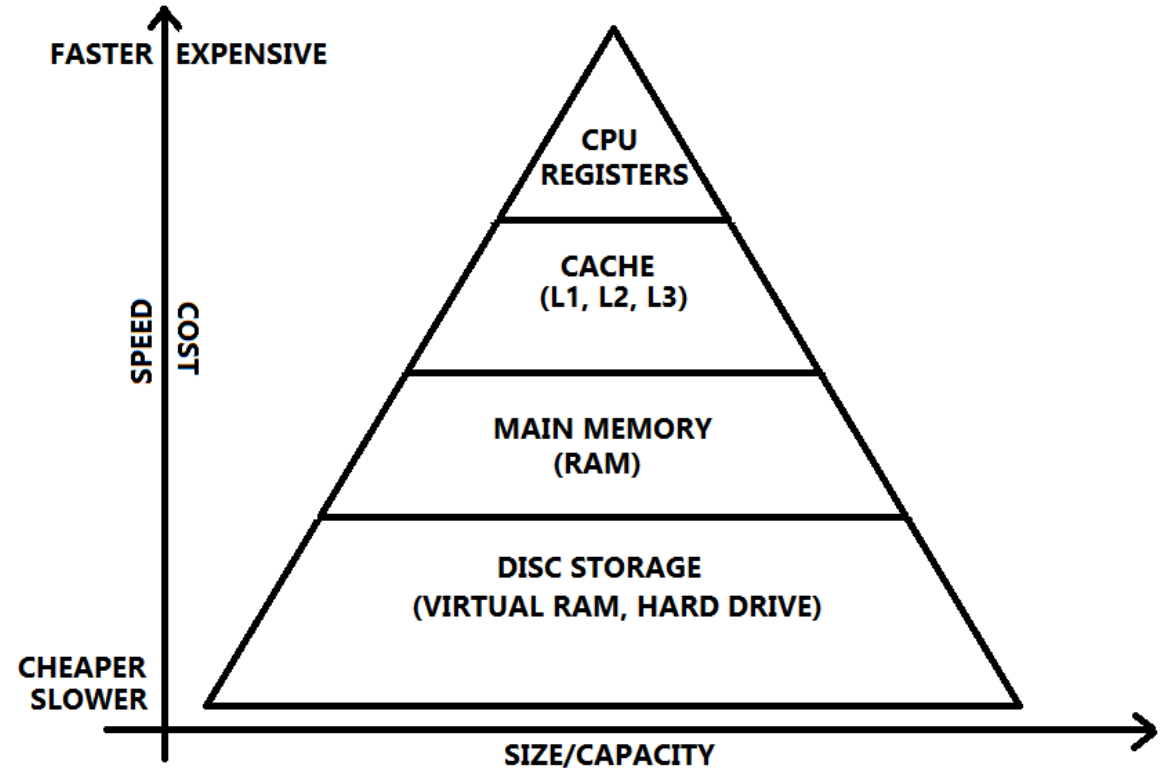
Architecture



- Instruction Set Architecture
- OS determines which data is loaded into memory and processed
- Instruction Set determines how data is loaded into memory and processed
- Most common Instruction Set Architectures scheme is MIPS.

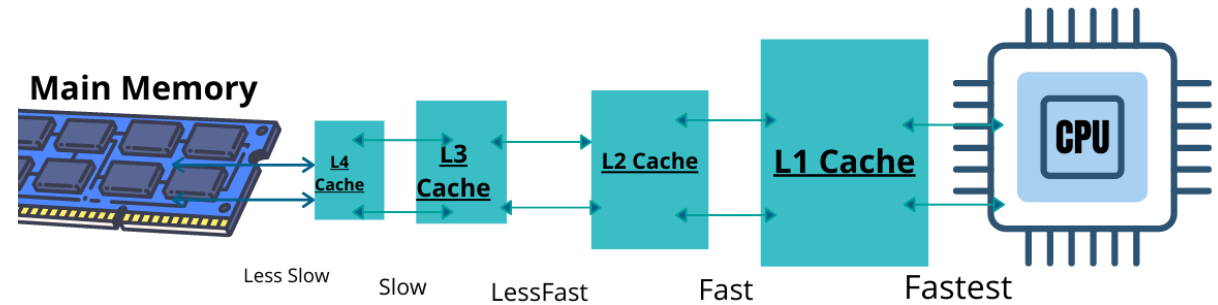
Types of Memory

- Levels of memory hierarchy
 - Disk Storage
 - Main Memory (RAM)
 - Cache
 - Registers
- Each level decreases in size, but increases in speed
- Cache (L1, L2) and registers are in the CPU



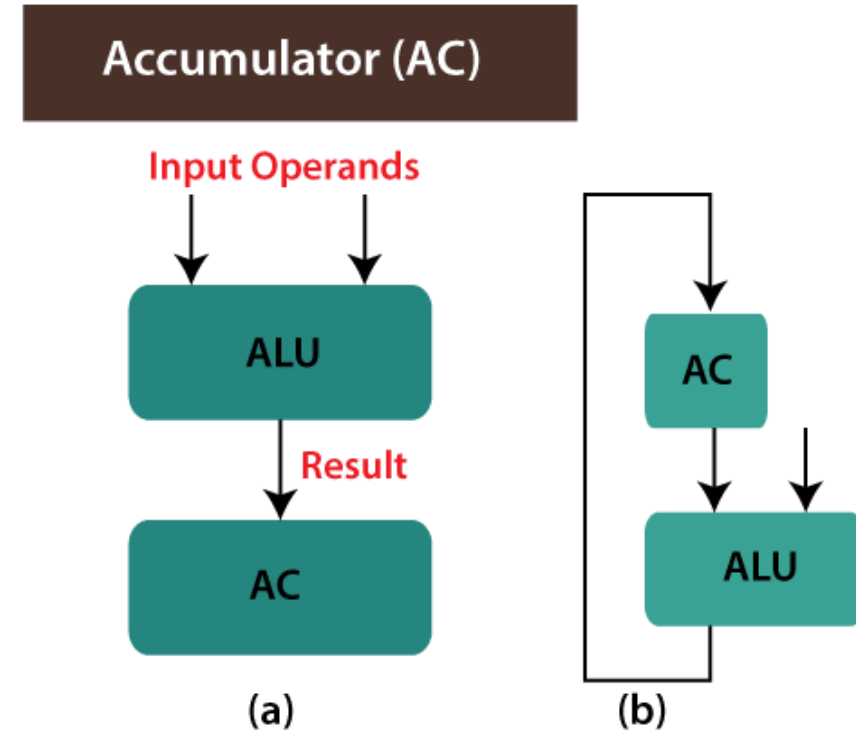
Cache

- Smaller and faster than Main Memory
- Located closer to a CPU core or ALU
- Stores frequently used data
- Purpose is to reduce average cost of memory fetch



Registers

- Smaller and faster than cache
- Fastest memory type
- Temporarily stores data to be used for program execution
- Located in CPU



AC is a register

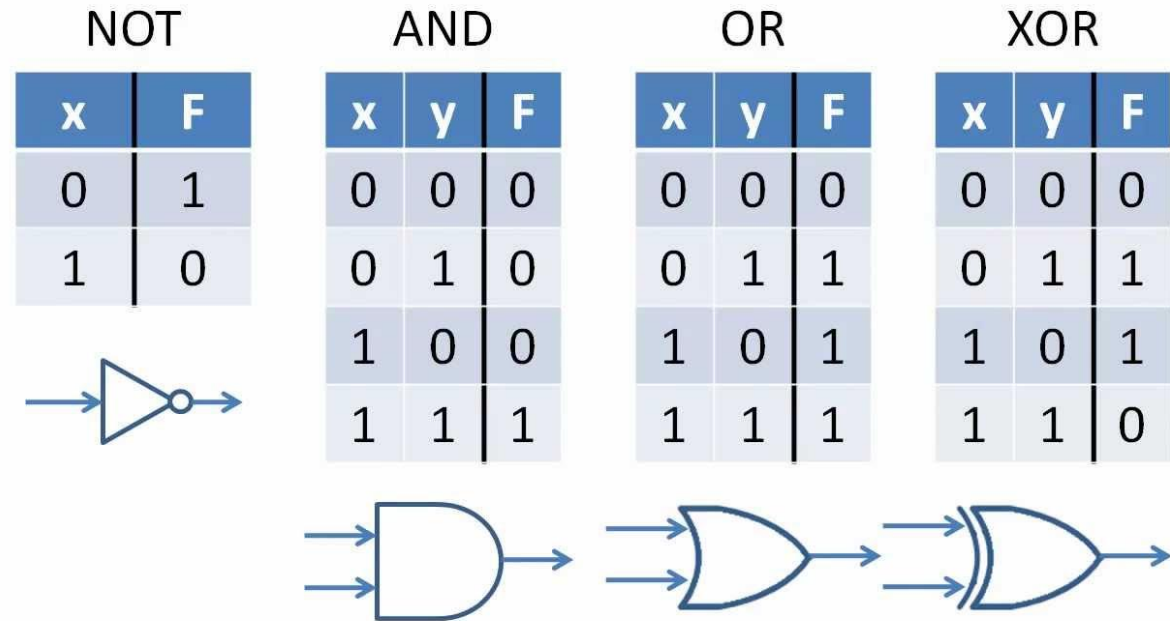
Memory Addressing

- Each memory cell has an address ID
- Each cell is located sequentially according to its address ID
- Address is in Hexadecimal format (base 16)
 - 0-9, A-F
 - Ex. 5 = 5 base 16, 10 = A base 16, 15 = F base 16
- All data is stored in binary format (0/1)

A d d r e s s e s	0xFFFFFFFF	1000 0000
		
		
	0x00000008	0100 1001
	0x00000007	1100 1100
	0x00000006	0110 1110
	0x00000005	0110 1110
	0x00000004	0000 0000
	0x00000003	0110 1011
	0x00000002	0101 0001
	0x00000001	1100 1001
	0x00000000	0100 1111
Main Memory		

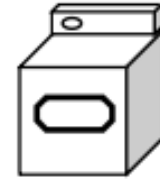
ALU

- Arithmetic Logic Unit (ALU)
- Performs following operations
 - Mathematical (+, -, /, *, etc.)
 - Logical (AND, OR, NOT, etc.)
- All operations are performed in binary (0/1)



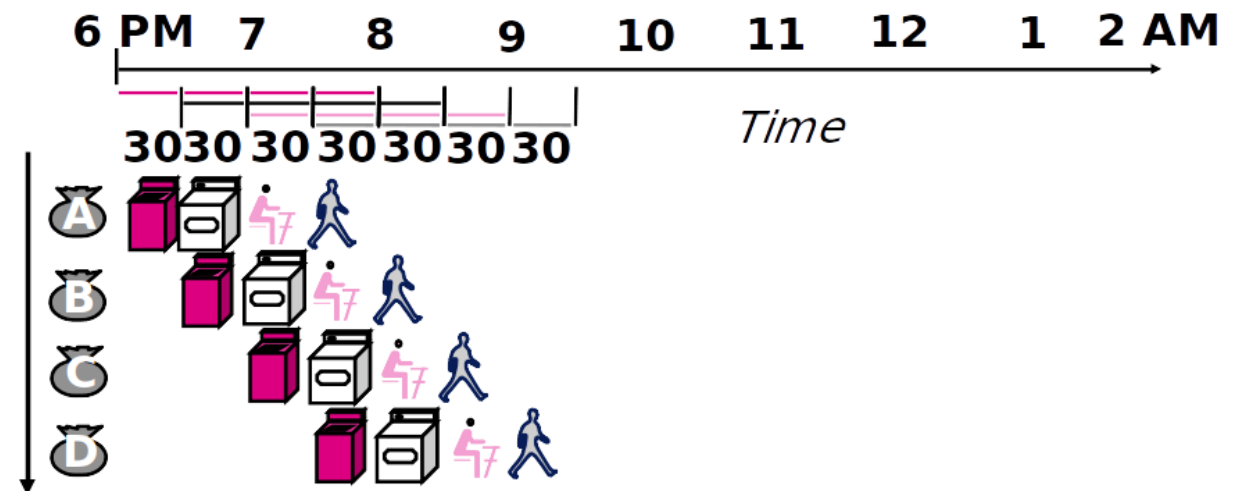
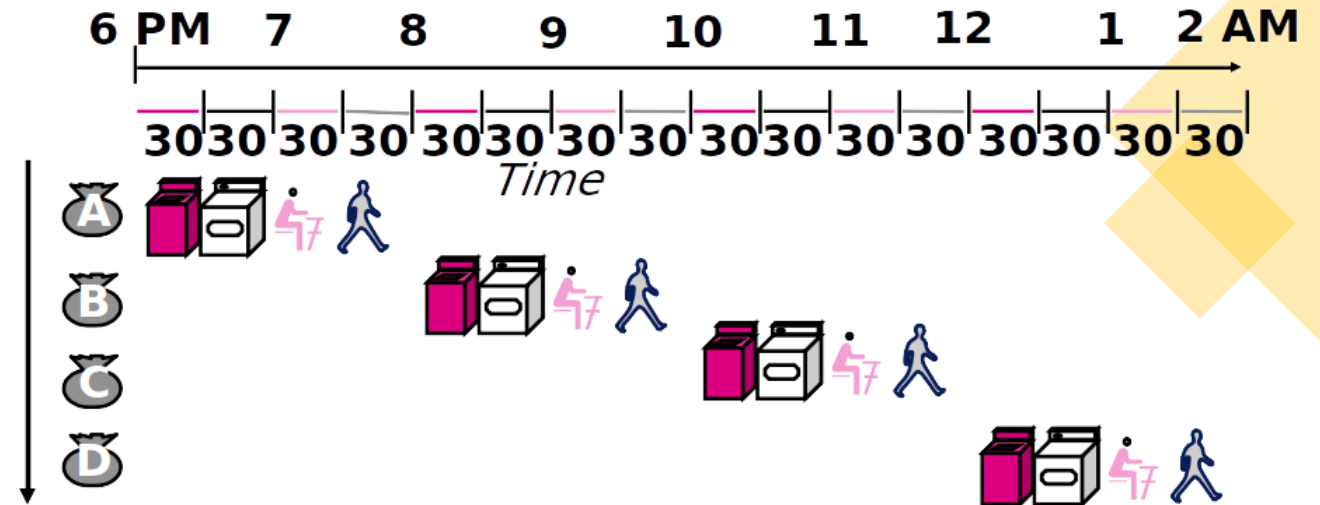
Instruction Execution

- Each instruction is made up of 5 steps
 - IF – Instruction fetched from memory
 - ID – Instruction decoded into register
 - EX – Instruction execution by ALU
 - MEM – Read/Write into memory
 - WB – Write back into register
- Think of them as steps to do laundry



Pipelining (Parallel Execution)

- Executing instructions sequentially takes up too much time and wastes resources.
- Pipelining is a strategy that takes advantage of available hardware resources by executing instructions in parallel



Assembly Language

- Assembly language is the lowest level of programming language
- Requires the use of memory addressing to write/read data from memory
- Requires user to specify register to write/read data from

LW R1, 0(0xFFFF)

LW R2, 5

Add R3, R1, R2

The above shows $(2 + 5)$, where 2 is fetched from memory address 0xFFFF

